AortaPlane: Parametric Learning for Aortic Annulus Plane Estimation in Computed Tomography Angiography

Vandhanaa Chitra[1,3], Emil Gasimov[1], Laura Bevis[1,2,5], Fuyu Cheng[5], Elisa Rauseo[4], Yousaf Bhatti[2,4], Hsu Hnin[1,3], Khushi Hiremath[1,6], Anthony Mathur[2,4], Caroline Roney[1,5], Gregory Slabaugh[1], and Xu Chen[1,2,*]

[1] Digital Environment Research Institute, Queen Mary University of London, UK
[2] NIHR Barts Biomedical Research Centre, Queen Mary University of London, Charterhouse Square, London, UK
[3] School of Electronic Engineering & Computer Science, Queen Mary University of London, UK
[4] Centre for Cardiovascular Medicine and Devices, William Harvey Research Institute, Queen Mary University of London, UK
[5] School of Engineering & Materials Science, Queen Mary University of London, UK
[6] School of Physical & Chemical Sciences, Queen Mary University of London, UK
xu.chen@qmul.ac.uk

Abstract. The precise identification of the plane of the aortic annulus from cardiac CT is an essential prelude to performing Transcatheter Aortic Valve Implantation (TAVI), currently done via manual annotation by radiologists. We propose an automated framework that estimates the aortic annulus plane via direct 6-DoF regression of the plane normal and centre from a 3D ResNet-CBAM encoder, augmented with CoordConv as our final model. We develop and evaluate this method on a cohort of 42 patients scanned under four different protocols, using an identical leak-free 5-fold cross-validation scheme. AortaPlane achieves a mean centre error of 9.36 mm and mean angular error of 11.33∘. We compare against a common existing approach, a U-Net-based landmark-heatmap model that identifies the 3 cusp landmarks and computes the plane from them: our method yields a statistically significant improvement in centre error (9.36 mm vs. 16.64 mm, Wilcoxon $p < 0.0001$), while the two do not differ significantly in angular accuracy ($p = 0.32$). Extending AortaPlane with the nnLandmark encoder forms our best-performing model, achieving 3.07 mm centre MAE and 13.08∘ angular error. We further demonstrate that disagreement between the predictions of the two independently trained models on the same subject constitutes a statistically significant yet leak-free measure of prediction failure ($\rho = 0.358$, $p = 0.0297$, $n = 37$), and formulate this as a Mondrian split-conformal gated review approach with an angular error probability bound. We additionally report a CoordConv ablation, a test-time augmentation robustness study, and a structural limitation of the 128^3-resolution pipeline.

Keywords: Aortic annulus detection, TAVI planning, Deep learning, Landmark localization

1 Introduction

Transcatheter Aortic Valve Implantation (TAVI) has become a minimally invasive treatment for severe aortic stenosis and an alternative to surgical aortic valve replacement, particularly in elderly patients, especially those aged over 75-80 years, who are often at increased surgical risk due to advanced age, frailty, and multiple comorbidities (Otto et al., 2021; Vahanian et al., 2022). Successful implantation of a single prosthetic valve and favourable procedural outcomes in TAVI are strongly influenced by appropriate prosthesis sizing and precise valve positioning [8]. Accurate localisation of the annular plane is therefore critical, as errors in prosthesis sizing or implantation depth may lead to clinically significant complications, including paravalvular regurgitation, coronary obstruction, and conduction disturbances, potentially necessitating additional interventions such as implantation of a second transcatheter valve, coronary ostial stenting, or permanent pacemaker implantation [4, 3, 8]. The annular plane is conventionally defined by the three basal hinge points of the right coronary (RC), non-coronary (NC), and left coronary (LC) cusps. In current clinical practice, this plane is identified manually by radiologists using multi-planar reconstructions of Computed Tomography Angiography (CTA) scans [5].This process is slow, requires specialised expertise, and shows measurable inter-observer disagreement: reported inter-observer variability for CT-derived annulus planning angles is $5.9° \pm 6.1°$ [14], a useful human baseline. Our model's angular error (11.33°) still exceeds this; a comparable human baseline for centre-point error is not established in the literature.

Deep learning provides a potential means of automating anatomical localisation in cardiac CTA. The existing methods show that the aortic annulus localisation predominantly follows a landmark- or segmentation-based paradigm [15, 10, 2, 6] where the RC, LC, and NC cusps are identified first before the plane is reconstructed geometrically. CTA is the modality of choice for this task because contrast enhancement renders the aortic lumen and cusp boundaries clearly visible, in contrast to non-contrast CT or echocardiography, where the annular anatomy is far less distinguishable.

An alternative is to formulate annulus-plane localisation as a direct regression problem, bypassing intermediate landmarks entirely. Comparing both paradigms head-to-head, as we do here, also enables using disagreement between the two independently trained models as a signal for flagging likely-incorrect predictions (Section 4).

This motivates an automated system that takes a 3D cardiac CT volume and predicts the annulus plane, parametrised by a unit normal vector and a centre point. Automated approaches to this problem fall into two broad strategies. The first treats the plane estimation as a direct regression task, which is an end-to-end network mapping the 3D volume to six numerical outputs representing the plane. The plane itself has five degrees of freedom (centre + normal direction), but we regress six values: three centre coordinates, three normal components, normalised to unit length avoiding the sign ambiguities and instability of minimal 5-parameter representations. The second treats it as a two-stage

landmark localisation task of detecting the three anatomical cusps individually and deriving the plane geometrically. To our knowledge, no published work runs a controlled, same-dataset, same-split ablation directly comparing these two paradigms for annulus-plane estimation; published systems are architecturally committed to the landmark/segmentation side of this choice [6, 2]. This is the central methodological gap this paper addresses. To the best of our knowledge, these paradigms have not been systematically compared for aortic annulus-plane localisation using identical patient data, preprocessing, cross-validation splits, and evaluation criteria. The principal contributions of this work are:

(1) We propose AortaPlane, an efficient encoder-only 3D deep learning framework that directly predicts the aortic annulus centre and plane orientation from cardiac CT using a parameter geometric representation, without requiring intermediate landmark localisation or segmentation.
(2) We conduct a controlled evaluation on a heterogeneous dataset comprising four CT acquisition protocols, comparing direct regression with a landmark-based approach under identical preprocessing, training, and five-fold cross-validation.
(3) We demonstrate that direct regression achieves significantly more accurate annulus centre localisation while maintaining comparable plane orientation accuracy, highlighting the complementary characteristics of the two paradigms.
(4) We show that disagreement between two independently trained models provides an effective indicator of prediction failure, enabling automatic quality assessment through a Mondrian split-conformal review framework.
(5) We show that substituting a self-configuring nnU-Net-derived encoder (nnLandmark) into AortaPlane yields our best-performing model, substantially improving centre localisation while remaining competitive in angular accuracy, with the parameter cost reported transparently.

2 Methods

We proposed *AortaPlane*, a direct-regression framework for estimating the aortic annulus centre and plane orientation and compared it with a matched landmark-based approach. Both models share the same 3D encoder backbone, differing only in output head and decoding strategy, so that any performance difference reflects the direct-regression vs. landmark-derived formulation rather than confounding architectural choices. We further evaluate a CoordConv . extension of AortaPlane designed to improve absolute spatial localisation.

Data The dataset comprises 42 patient scans across four cohorts. D (Dongyang), K (KiTS), and R (Rider) are subsets of the AVT dataset [17], a multicentre collection of 56 full-body angiogram CTs with ground-truth aortic segmentations; our team manually annotated RC, NC, and LC annular landmarks on a suitable subset of these scans in which the aortic root was sufficiently visible, using multi-planar reconstruction in 3D Slicer and following standard clinical protocol (n=XX) [5]. The fourth cohort differed in image intensity representation

and preprocessing requirements. CONTCT_R (20 patients, standard HU CT), was curated from retrospective TAVI cases at Barts Health NHS Trust, with segmentations and landmarks manually created from the full-body angiograms. Each patient provides a CT volume, a segmentation mask, and a landmark file. K requires percentile clipping (12-bit, non-HU); R is HU-valued but needs special handling for −2000 background padding. Five patients, covering all four cohorts (two from CONTCT_R, one each from D, K, R [16]), were held out for testing and never used in training or tuning. The remaining 37 were used for 5-fold cross-validation.

2.1 AortaPlane

AortaPlane is a 3D ResNet-18-like encoder [9] with CBAM attention [20] added after each residual layer, followed by a two-layer regression head giving six outputs [nx, ny, nz, cx, cy, cz] directly, the plane's centre-and-normal (c, n) form, which is the representation used throughout training and evaluation for both AortaPlane and Baseline. The CBAM variant was selected via a preliminary encoder ablation over standard ResNet-18, squeeze-and-excitation attention, and CBAM. Section 3.2 extends this encoder with CoordConv to obtain our final model, whose mean and median angular error are also reported there. AortaPlane is trained end-to-end with a loss combining angular alignment and centre regression:

$$\mathcal{L}_{AortaPlane} = \lambda_n \left(1 - |\hat{\mathbf{n}} \cdot \mathbf{n}|\right) + \lambda_c \|\hat{\mathbf{c}} - \mathbf{c}\|_2^2 \tag{1}$$

Baseline: Landmark / U-Net. Baseline model has the same ResNet-CBAM encoder architecture, but a U-Net-like decoder [18, 7] is added along with three independent 1×1×1 convolutional heads per cusp landmark which give three heatmaps that are turned into coordinates using soft-argmax. The plane is calculated geometrically in the post-processing step as

$$\hat{\mathbf{c}} = \frac{\hat{\mathbf{x}}_{RC} + \hat{\mathbf{x}}_{NC} + \hat{\mathbf{x}}_{LC}}{3}, \qquad \hat{\mathbf{n}} = \text{normalise}\big((\hat{\mathbf{x}}_{NC} - \hat{\mathbf{x}}_{RC}) \times (\hat{\mathbf{x}}_{LC} - \hat{\mathbf{x}}_{RC})\big) \tag{2}$$

where $\hat{\mathbf{x}}_{RC}, \hat{\mathbf{x}}_{NC}$, and $\hat{\mathbf{x}}_{LC}$ represent the estimated RC, NC and LC points respectively. The loss combines the same angular and centre terms (computed on the derived plane, enabling direct comparison with AortaPlane) with a spread regularisation term, $\mathcal{L}_{spread} = \max(0, m - \min_{i \neq j} \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|_2)$, penalising any pair of predicted landmarks closer than margin m and preventing degenerate collapse onto a single point. We note that this loss supervises the derived plane (centre and normal) rather than the three landmark coordinates directly, modulo the spread term preventing collapse, individual RC/NC/LC predictions are only weakly constrained so long as their centroid and cross-product recover the correct plane. A direct per-point MSE term on the three landmark coordinates is a natural extension not evaluated here due to time constraints, and is flagged as future work (Section 7).

2.2 CoordConv Extension

Standard convolution is translation invariant, which is known to be limiting for spatial localisation tasks [13], as in centre-point regression. For an input volume $X \in R^{D\times H\times W}$, we construct three coordinate channels

$$C_d[d,h,w] = \frac{2d}{D-1} - 1, \quad C_h[d,h,w] = \frac{2h}{H-1} - 1, \quad C_w[d,h,w] = \frac{2w}{W-1} - 1, \tag{3}$$

each normalised to $[-1, 1]$, and concatenate them channel-wise with X before the first convolutional layer: $X' = \text{concat}(X, C_d, C_h, C_w)$. BaselineAll subsequent network components remain unchanged, and CoordConv is disabled in the CBAM-only (pre-CoordConv) configuration of AortaPlane. The overhead is negligible: only the first layer's input channel count changes, adding 3 channels out of 3.62M total parameters versus 3.59M for the CBAM-only variant

Training and Cross-Validation. Both models are trained identically using 5-fold cross-validation on the 37 training patients (the five test patients excluded from every fold), yielding five checkpoints per model per fold. Training uses the Adam optimiser with automatic mixed precision, early stopping (patience 60, up to 300 epochs), on the QMUL HPC cluster [12] using a single GPU node. At inference, each individual test patient's five fold checkpoints are ensembled: the mean gives the point estimate, and the standard deviation gives the uncertainty signal.

Preprocessing Pipeline. Each volume is resampled to 1.0 mm isotropic resolution using the SimpleITK library; landmark coordinates are converted from image-voxel to physical (world, mm) coordinates and back to target-volume voxel coordinates via explicit LPS/RAS handling. Ground-truth centre is the mean of the three cusp points; ground-truth normal is the cross product of the two cusp-to-cusp vectors. Intensities are clipped per cohort, HU windowing for CONTCT_R, D, and R series, and percentile clipping for the 12-bit K series, since a single HU window saturated 50–80% of K-series voxels. A fixed 128×128×128 mm field of view is cropped around the annulus, centred at inference on the ground-truth annulus centroid, and resized to 64^3 (2.0 mm/voxel) or 128^3 (1.0 mm/voxel); this dual-resolution scheme trades off spatial fidelity against the memory and compute cost of 3D convolution. Intensities are normalised to [0, 1]. We emphasise that because the crop is centred on the ground-truth centroid, the task evaluated here is annulus-plane estimation within a pre-localised aortic-root ROI, not fully automatic annulus localisation from an unconstrained CT volume; a genuinely automatic pipeline would require an upstream coarse-localisation stage (e.g. a jittered or segmentation-derived crop centre) to identify this ROI without ground truth, which we flag as a structural limitation left to future work (Section 7).

Table 1. Headline comparison of AortaPlane, Baseline, and AortaPlane+nnLandmark (Final Model) under the identical 5-fold cross-validation protocol ($n = 37$; each patient scored on its single held-out fold).

Model	Centre (mm)	MAE (mm)	MSE (mm^2)	Angle (°)	$\leq 10°$
AortaPlane (direct)	9.36	4.84	40.10	11.33	18/37
Baseline (landmark)	16.65	8.12	115.01	13.27	16/37
AortaPlane+nnL.	6.64	3.07	26.86	13.08	16/37

2.3 nnLandmark Backbone Integration

To test whether a self-configuring segmentation architecture outperforms our hand-designed encoders, we substitute nnU-Net's `PlainConvEncoder`/`PlainConvUNet` [11] (InstanceNorm3d normalisation, LeakyReLU activations, channel widths 32–320) in place of our native ResNet-CBAM encoder in both AortaPlane and Baseline, which we refer to as *nnLandmark*. Task formulation, loss functions, and training protocol are kept identical to the native models; only the encoder is replaced. This isolates any performance difference to the choice of encoder alone. The AortaPlane+nnLandmark and Baseline+nnLandmark variants are trained using the same 5-fold cross-validation procedure on the same 37-patient training set as the native models.

3 Results

Overall Performance. Table 1 compares AortaPlane (direct regression), Baseline (landmark-derived), and our final model (AortaPlane augmented with nnLandmark's backbone), all evaluated under the identical 5-fold cross-validation protocol ($n = 37$; each patient scored on the single fold in which it was held out). AortaPlane+nnLandmark wins decisively on every centre-based metric, while its angular error sits between AortaPlane's and Baseline's, matching Baseline's $\leq 10°$ count rather than exceeding AortaPlane's. Qualitative examples comparing ground truth, AortaPlane, and Baseline predictions are shown in Fig. 1.

nnLandmark on the True Test Set. Unlike Table 1, which pools CV-fold predictions across all 37 patients, evaluating instead only on the 5 patients held out entirely from training including native baselines explains why the native AortaPlane/Baseline figures above differ numerically from Table 1 despite being identical trained models (full backbone ablation in Table 2).

Swapping AortaPlane's encoder for nnLandmark (Section 2.3) yields our final model, improving both axes simultaneously: angular error falls from 20.83° and centre error from 6.16 mm to 3.17 mm a clean win, not a trade-off. The same substitution in Baseline is more mixed: centre error improves ($5.70 \rightarrow 3.91$ mm) but angular error worsens ($17.45° \rightarrow 22.65°$). Both nnLandmark variants' angular errors concentrate on the same two test patients, whose scans fall outside the distribution its default configuration was tuned for; we target this gap in future work.

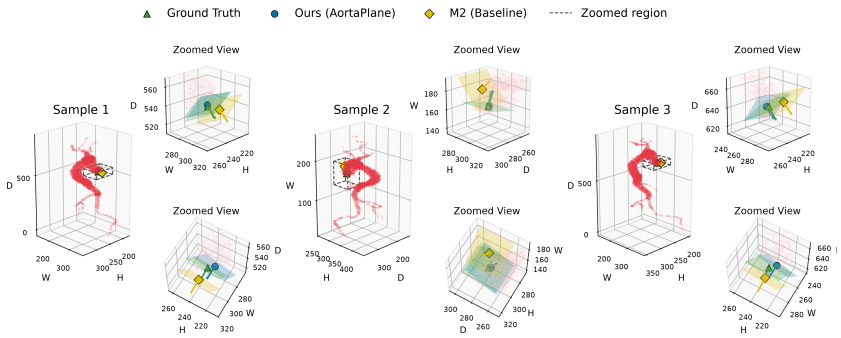


Fig. 1. Ground Truth, AortaPlane, and Baseline for three patients (Sample 1, Sample 2, Sample 3), each with overview and zoomed views. Sample 2 shows Baseline's angular error at 47.00°, while AortaPlane's angular error is 5.26°.

Table 2. Backbone/encoder ablation. AortaPlane and Baseline, native and with each alternative backbone substituted, evaluated on the 5 true held-out test patients.

Variant	Centre MAE (mm)	Angle MAE (°)	≤ 10	Params
AortaPlane (native)	6.16±2.86	20.83±16.41	1/5	3.62M
AortaPlane + U-Net	7.14±4.24	17.60±7.55	1/5	5.90M
AortaPlane + DenseNet3D	4.14±2.29	17.82±11.72	1/5	0.43M
AortaPlane + nnLandmark	**3.17±2.61**	**15.79±10.96**	**3/5**	**88.21M**
Baseline (native)	5.70±2.95	17.45±7.66	1/5	5.89M
Baseline + ResNet3D (no skip)	3.54±3.61	20.25±12.96	0/5	5.31M
Baseline + DenseNet3D	5.13±2.70	21.53±13.41	1/5	0.99M
Baseline + nnLandmark	3.91±2.87	22.65±9.92	0/5	88.62M

We additionally report three component ablations on native AortaPlane (n = 37, pooled 5-fold cross-validation). Increasing input resolution from 64^3 to 128^3 reduces angular error by 43%, confirming that finer voxel resolution meaningfully improves plane orientation accuracy. Test-time augmentation (TTA) yields an asymmetric effect: centre error improves significantly ($p = 0.0013$), while angular error worsens significantly ($p = 0.0275$), suggesting TTA trades off orientation stability for centroid precision. Finally, the CoordConv3D extension shows no significant effect on either metric ($p = 0.625$), a null result we report honestly rather than omit.

Cross-Model Disagreement as a Deployment-Ready Gating Signal. A clinically relevant question is whether the system can flag likely-wrong predictions at inference time, without ground truth. **Within-model fold disagreement** the std. dev. of a model's own 5-fold CV predictions correlates poorly with its own error ($\rho = -0.100$, $p = 0.87$ for AortaPlane; $\rho = 0.800$, $p = 0.10$, n.s. at $n = 5$, for Baseline), missing one large-error patient entirely. **Cross-model disagreement**, using only each patient's genuinely held-out fold, instead

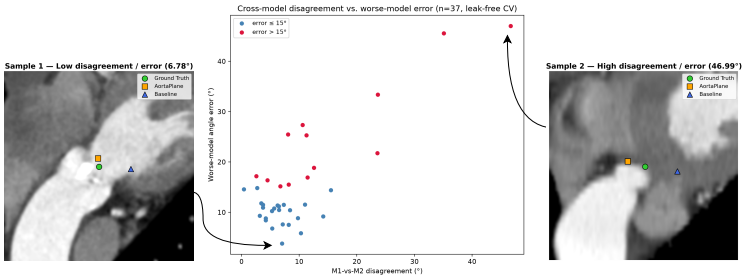


Fig. 2. Cross-model disagreement vs. worse-model error across all 37 CV patients (centre), flanked by a representative low-disagreement case (Sample 1, left) where GT/AortaPlane/Baseline cluster tightly, and the highest-disagreement case (Sample 2, right) where Baseline fails while AortaPlane remains accurate. Red points ($> 15°$ error) cluster at higher disagreement, consistent with $\rho = 0.358$.

shows a modest correlation with the *worse*-performing model's error ($\rho = 0.358$, $p = 0.0297$) (Fig. 2), but not with AortaPlane's own error specifically ($\rho = -0.124$, $p = 0.47$, n.s.); at inference time only disagreement is observable, not which model is worse. Part of this correlation follows near-mechanically from a geometric lower bound ($\max(e_1, e_2) \geq$ disagreement/2), so we do not claim disagreement predicts AortaPlane's own error rather, we use it as a symmetric trigger for expert review whenever the two models disagree substantially. Two cases illustrate this bidirectionality: Sample 2's largest disagreement (46.67°) reflects a near-catastrophic Baseline failure (47.00°) while AortaPlane stays accurate (5.26°); another patient (35.07° disagreement) shows the reverse (AortaPlane 45.57° vs. Baseline 17.44°). **Threshold sweep.** Across three "bad"-prediction thresholds ($> 10°$, $> 15°$, $> 20°$, $n = 37$), corresponding disagreement cut-points of 0.40°, 7.96°, and 15.51° give precision/recall/F1 of 0.75/0.96/0.84, 0.76/0.77/0.76, and 1.00/0.57/0.73 respectively descriptive rather than a generalisation guarantee, which the calibrated bound below addresses.

Conformal calibration. We formalise this via Mondrian split-conformal calibration [19, 1], conditioning on the 15° disagreement group, to bound the *worse*-performing model's error rather than to identify which model is worse. At 90% coverage, the low-disagreement group ($n = 32$) achieves a calibrated bound of $\leq 19.08°$; the high-disagreement group ($n = 5$) yields $\leq 45.57°$ at 80% coverage (90% is not achievable at this sample size). Leave-one-out coverage matched nominal targets throughout. This converts a correlational, group-level finding into a calibrated guarantee for flagging cases for expert review using only quantities available at inference time without requiring the determination of which specific model is correct.

Across three candidate 'bad'-prediction definitions, the corresponding disagreement thresholds were 0.40°, 7.96°, and 15.51° for error thresholds of 10°, 15°, and 20° respectively. At the 10° threshold, 36 of 37 patients were flagged, with 27 true positives (precision 0.75, recall 0.96, F1 0.84). At 15°, 17 patients

were flagged with 13 true positives (precision 0.76, recall 0.77, F1 0.76). At the strictest 20° threshold, only 4 patients were flagged, all true positives (precision 1.00, recall 0.57, F1 0.73), reflecting a conservative but perfectly precise regime at high disagreement.

4 Conclusion and Future Work

We presented a controlled, same-dataset comparison of direct plane regression and landmark-derived estimation for aortic annulus localisation from cardiac CT. Direct regression significantly improved centre localisation while achieving equivalent angular accuracy. Extending AortaPlane with the nnU-Net-derived nnLandmark encoder produced the best overall performance. Cross-model disagreement further provided a calibrated trigger for expert review through Mondrian split-conformal gating, offering a practical safeguard before predictions inform TAVI planning.

Future work will address the 128^3 crop-anchoring limitation through automatic coarse localisation, extend CoordConv and direct point-wise supervision to Baseline, benchmark against external methods [6], and prospectively validate disagreement gating and per-centre generalisability in a larger independent multi-centre cohort.

Acknowledgements

This work forms part of the research portfolio of the National Institute for Health and Care Research (NIHR) Barts Biomedical Research Centre (NIHR203330).

References

1. Angelopoulos, A.N., Bates, S.: A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv preprint (2021)
2. Astudillo, P., Mortier, P., Bosmans, J., Backer, O.D., de Jaegere, P., Iannaccone, F., Beule, M.D., Dambre, J.: Automatic detection of the aortic annular plane and coronary ostia from multidetector computed tomography. Journal of Interventional Cardiology **2020**, 9843275 (2020)
3. Bhushan, S., Huang, X., Li, Y., He, S., Mao, L., Hong, W., Xiao, Z.: Paravalvular leak after transcatheter aortic valve implantation: Its incidence, diagnosis, clinical implications, prevention, management, and future perspectives — a review article. Current Problems in Cardiology **47**(10), 100957 (2022)
4. Bianchini, F., Bianchini, E., Romagnoli, E., Aurigemma, C., Zito, A., Busco, M., Nesta, M., Bruno, P., Laezza, D., Giambusso, N., Natale, L., Pelargonio, G., Burzotta, F., Trani, C.: Anatomical annulus predictors of new permanent pacemaker implantation risk after balloon-expandable transcatheter aortic valve implantation. American Journal of Cardiology **224**, 26–35 (2024)
5. Blanke, P., Weir-McCall, J.R., Achenbach, S., Delgado, V., Hausleiter, J., Jilaihawi, H., Marwan, M., Nørgaard, B.L., Piazza, N., Schoenhagen, P., Leipsic, J.A.: Computed tomography imaging in the context of transcatheter aortic valve implantation

(tavi)/transcatheter aortic valve replacement (tavr): An expert consensus document of the society of cardiovascular computed tomography. JACC: Cardiovascular Imaging **12**(1), 1–24 (2019)
6. Cho, Y., Park, S., Hwang, S.H., Ko, M., Lim, D.S., Yu, C.W., Park, S.M., Kim, M.N., Oh, Y.W., Yang, G.: Aortic annulus detection based on deep learning for transcatheter aortic valve replacement using cardiac computed tomography. Journal of Korean Medical Science **38**(37), e306 (2023)
7. Çiçek, Ö., Abdulkadir, A., Lienkamp, S.S., Brox, T., Ronneberger, O.: 3d u-net: Learning dense volumetric segmentation from sparse annotation. In: MICCAI. pp. 424–432 (2016)
8. Généreux, P., Piazza, N., Alu, M.C., et al.: Valve academic research consortium 3: Updated endpoint definitions for aortic valve clinical research. European Heart Journal **42**(19) (2021)
9. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of CVPR. pp. 770–778 (2016)
10. Ionasec, R.I., Voigt, I., Georgescu, B., Wang, Y., Houle, H., Vega-Higuera, F., et al.: Patient-specific modeling and quantification of the aortic and mitral valves from 4-d cardiac ct and tee. IEEE Transactions on Medical Imaging **29**(9), 1636–1651 (2010)
11. Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H.: nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods **18** (2021)
12. King, T., Butcher, S., Zalewski, L.: Apocrita — high performance computing cluster for queen mary university of london (2017)
13. Liu, R., Lehman, J., Molino, P., Such, F.P., Frank, E., Sergeev, A., Yosinski, J.: An intriguing failing of convolutional neural networks and the CoordConv solution. In: Advances in Neural Information Processing Systems. vol. 31, pp. 9605–9616 (2018)
14. Mehier, B., Dubourg, B., Eltchaninoff, H., Durand, E., Tron, C., Cribier, A., Michelin, P., Dacher, J.N.: Mdct planning of trans catheter aortic valve implantation (tavi): determination of optimal c-arm angulation. International Journal of Cardiovascular Imaging **36**(8) (2020)
15. Palmer, R.I., Xie, X., Tam, G.K.L.: Automatic aortic root segmentation with shape constraints and mesh regularisation. In: Proceedings of BMVC. Swansea, UK (2015)
16. Pepe, A., Melito, G.M., Egger, J. (eds.): Segmentation of the Aorta. Towards the Automatic Segmentation, Modeling, and Meshing of the Aortic Vessel Tree from Multicenter Acquisition: First Challenge, Lecture Notes in Computer Science, vol. 14539. Springer, Cham (2024)
17. Radl, L., Jin, Y., Pepe, A., Li, J., Gsaxner, C., Zhao, F.h., Egger, J.: Avt: Multicenter aortic vessel tree cta dataset collection with ground truth segmentation masks. Data in brief **40**, 107801 (2022)
18. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: MICCAI. pp. 234–241 (2015)
19. Vovk, V., Gammerman, A., Shafer, G.: Algorithmic Learning in a Random World. Springer, New York (2005)
20. Woo, S., Park, J., Lee, J.Y., Kweon, I.S.: CBAM: Convolutional block attention module. In: Proceedings of ECCV. pp. 3–19 (2018)